



March 2015

Team: SC, RM, RS, LC, NW (50%)

Strategy

- [RS] Structured Credit Monitor: Work-in-progress. Baseline on J.P. Morgan Structured Credit Report and identify key sectors and spreads. Developed preliminary report based on these sectors/spreads.
- [LC] Research and reporting on landscape of primary mortgage market
- [SC] Research brief on Mortgage Credit Availability
- [SC] Presentation: Role of HFs in a Fixed Income portfolio
- [RS] Surveillance reports for GSE loan-level data should be completed by mid-April. Significant work was required in terms of identifying modified loans.

Modeling

- **Scale:** Reorganize and refactor existing CCMQuantlib fixed income analytics library (written by Bin Shao) as Scale. See the Scale presentation (distributed separately) for an overarching view of the goals of this library.
 - [LC] Productionized 30-year FNMA and GNMA current coupon calculation
 - [JH, NW] Swap Yield Curve: Productionize daily Swap Curve construction. Save forwards, spot curves, par yields to the database on a daily basis. Load history (NW).
 - [JH, NW] New Scale: Work on productionizing Quantlib code that replicates Bin's code (current focus: yield curve construction). If successful, this approach would provide an excellent platform for our Scale project. Quantlib is:
 - Reasonably well-tested: it's been around since 2000
 - Well-maintained: There are at least a couple of releases every year
 - Rich feature set: Quantlib functionality includes a significant collection of utilities (Date libraries, optimization engines, splines etc.), yield curve builders, interest rate models etc.
 - Documented code
- [RM] Accordion: Roll-rate model for GSE loan-level data
 - Model specification and estimation completed



- Current focus: Forecasting and model tracking